

RESEARCH ARTICLE

Design, Development, and Field Testing of a Tomato Bunch Harvesting Robot

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Received: 28 May 2025 | **Revised:** 20 December 2025 | **Accepted:** 21 May 2026

Funding: Fruit and Vegetable Harvesting Robot Development, Grant/Award Number: Yue Cai [2023] 6; Fundamental Research Funds of the Central Universities, Grant/Award Number: 2024ZYGXZR028

Keywords: agricultural robots | end effector | tomato bunch and stalk recognition and localization | tomato harvesting

ABSTRACT

With the aging population and labor shortages, the proportion of labor costs in tomato harvesting is increasing, making the development of tomato harvesting robots imperative. This study developed an integrated tomato bunch harvesting robotic system for cherry tomatoes. A combined cutting and gripping end-effector powered by a single actuator, achieving a cutting success rate of 93.33% and a gripping capacity of 1600 g. A parameterized camera arrangement was employed to match the robotic arm's field of view, thereby avoiding mutual interference. A tomato bunch and stalk recognition model was constructed based on the YOLOv4 algorithm to enable precise localization of harvesting points. The proposed tomato bunch-stalk matching method achieved a recall rate of 99.22%, while the stalk growth-direction qualitative discrimination method attained an accuracy of 97%. Field experiments demonstrated that the system achieved an average harvesting time of 12.23 s per tomato bunch and an overall harvesting success rate of 70.77% in unstructured environments, improving automation and operational efficiency compared to existing solutions. This research offers a solution integrating hardware optimization and perception algorithms for greenhouse harvesting robots, demonstrating potential for commercial application.

1 | Introduction

China is the largest country in terms of protected agriculture cultivation area worldwide. Among greenhouse-grown crops, tomatoes hold significant economic importance, with cherry tomatoes becoming a key cultivated variety due to their high added value (Xiao et al. 2024). Currently, the harvesting of cherry tomatoes heavily relies on manual labor, with labor costs accounting for 30–50% of the total, and the harvesting process constituting 35% to 45% of the overall labor input (Vougioukas 2019; Iloh et al. 2020; Xiao et al. 2024; Zhang et al. 2024b). Therefore, it is urgent to develop tomato harvesting robots to reduce labor costs.

Although research on tomato harvesting robots began in Japan in the 1980s (Monta et al. 1998; Jizhan 2017; Rajendran et al. 2024), significant technical bottlenecks remain after decades of development. Early systems, such as the Mecanum wheel chassis combined with a UR5 robotic arm proposed by Yaguchi et al., achieved a single fruit harvesting time of 23 s but had a success rate of only 60% (Yaguchi et al. 2016). Subsequent studies, such as the rail-based robot by Feng et al., improved the success rate to 83.9%, with an efficiency of 24 s per fruit (Feng et al. 2015). However, further improvements were limited in automation due to reliance on manual track switching (Feng et al. 2018). The granular harvesting robot by Gao et al. reduced the single fruit harvesting time to 6.4 s but lacked fruit

collection functionality, resulting in insufficient system integrity (Gao et al. 2022). These typical solutions reveal three major technical challenges: difficulty in balancing harvesting efficiency and success rate, limited operational continuity, and insufficient integration of functional modules.

In the field of fruit perception technology, traditional image-based methods rely on features such as color (Zhuang et al. 2019), shape (Xiong et al. 2018), texture (Habegger et al. 2021), and multi-feature fusion (Xu et al. 2016). While these methods perform well in controlled environments, they struggle to handle the variable lighting and complex backgrounds present in agricultural settings (Hua et al. 2023). The introduction of deep learning techniques has significantly enhanced robustness (Saedi and Khosravi 2020; Lamb and Chuah 2018; Sun et al. 2018; Wang et al. 2018; Zhang et al. 2018). Among these, regression-based single-stage detection algorithms, such as YOLO and SSD, have become mainstream due to their real-time performance advantages (Liu et al. 2020; Peng et al. 2022; Zheng et al. 2022; Zhao et al. 2023). However, existing algorithms still require improvements in detecting small objects (such as stalks) and in densely occluded scenes, which directly limits the precise localization of harvesting points. This study adopts the YOLO framework as a foundation and applies post-processing to the recognition results, accurately identifying tomato stalk harvesting points and estimating the growth orientation of stalks, thereby overcoming the bottleneck in stalk growth orientation recognition.

In terms of end-effector design, existing solutions exhibit a contradiction between singular functionality and structural redundancy. The laser cutting approach (Liu et al. 2008) enables non-destructive separation but features a bulky device prone to collisions. Pneumatic drive systems (Feng et al. 2018) rely on external air sources, increasing system complexity. The latest compact designs, such as the cam-driven scissor mechanism by Rong et al., demonstrate integration potential (Rong et al. 2024). However, using a spring as the source of cutting force may result in insufficient cutting strength, and the performance of the end-effector was not discussed. Building upon this, we propose an innovative integrated grasping and cutting mechanism that achieves synchronized control of grasping and cutting through a single power source, featuring a compact structure and simple operation.

To address these issues, this study proposes a highly integrated tomato harvesting robot that combines a mobile chassis, robotic arm, end-effector, camera, and control system, enabling autonomous navigation, recognition, harvesting, and basket placement within tomato fields. The detailed descriptions of “Mobile chassis and autonomous navigation design” and “Motion planning for the robotic arm and end-effector” are provided in the *Supporting Information* file. The robot is capable of continuous tomato recognition and harvesting, achieving an overall success rate of 70.77%. The main contributions of this paper are as follows:

1. An integrated end-effector is developed for tomato bunch harvesting, combining gripping and cutting functions within a single unit. The scissor blades adopt a curved profile to prevent stalk slippage during harvesting. The

design achieves a cutting success rate of 93.33% and securely grips bunches weighing up to 1600 g, ensuring reliable stalk severing and stable bunch retention.

2. A parameterized method for camera position and orientation is proposed to align the camera's field of view with the robotic arm's working range. This approach increases the number of harvestable tomatoes within the camera's view by 16.43%.
3. A tomato stalk-to-bunch matching algorithm and a stalk growth-direction qualitative discrimination method are proposed. These approaches achieve a stalk-to-bunch matching recall of 99.22% and a growth-direction recognition accuracy of 97.44%. The developed tomato harvesting robot further attains an overall field harvesting success rate of 70.77%.

This paper presents the design and development of a tomato bunch harvesting robot tailored for cherry tomatoes, detailing the design of the end-effector, tomato recognition and localization, and the control system. The remaining sections of this paper are organized as follows. Section 2 introduces the tomato cultivation environment. Section 3 describes the hardware, control system, and perception system of the tomato harvesting robot. Section 4 presents experiments and discussions on deep learning-based recognition methods, stalk-to-bunch matching algorithms, the stalk growth-direction qualitative discrimination method, end-effector performance, and actual harvesting. Finally, Section 5 provides conclusions and suggestions for future improvements.

2 | Environment and Crop Information

This study focuses on cherry tomatoes cultivated in standard greenhouses, where diffused glass or film materials are used to ensure uniform lighting without direct sunlight. The tomatoes are primarily grown using soilless cultivation methods, as illustrated in Figure 1.

Standard greenhouses are typically divided into pedestrian areas and planting zones. The planting zones contain structures such as planting racks, drip irrigation systems, coconut coir strips, and tracks. The center distance between adjacent planting racks is 1600 mm, the lowest point height is 660 mm, and the spacing between hooks is 900 mm. The tracks, welded from steel pipes, accommodate equipment like spraying and irrigation devices and elevated operation vehicles. The pedestrian areas are generally flat, facilitating the transport of materials and the maneuvering of equipment between rows.

Typically, the leaves beneath maturing tomatoes are removed in advance, as they consume excessive nutrients and block sunlight, delaying the ripening process. During manual harvesting, scissors are used to cut the stalks of ripe tomato bunches, which are then collected to complete the harvesting process. After manual harvesting, the hang lines are loosened, and the stems of the harvested tomatoes are wound around hooks on the planting racks. This prevents the tomato tops from exceeding the greenhouse height and maintains the stalk height of maturing tomatoes between 1100 and 1500 mm, facilitating subsequent harvesting.

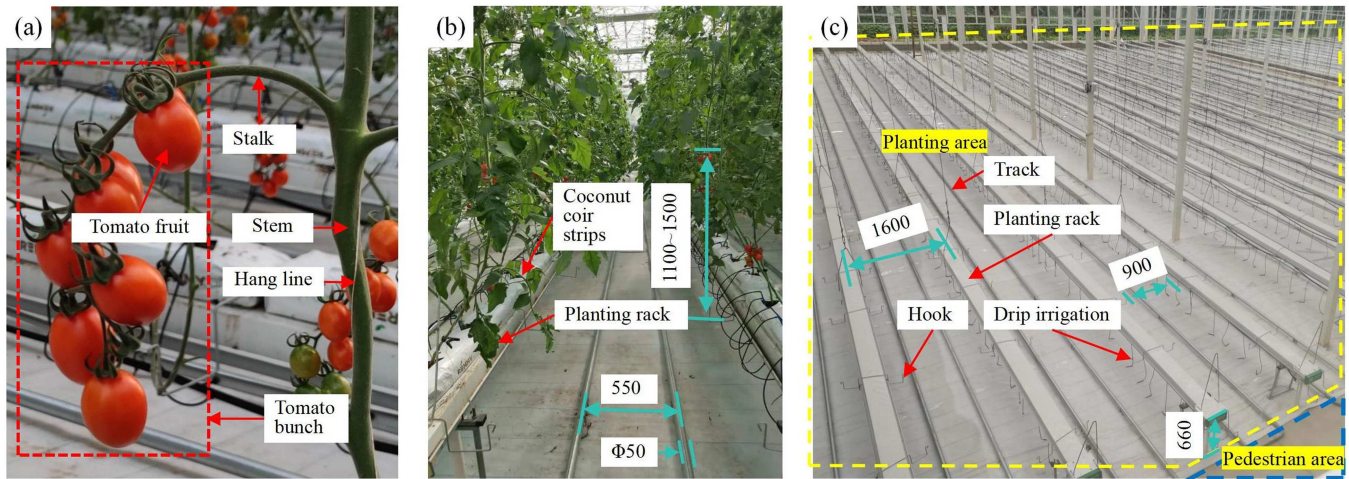


FIGURE 1 | Cherry tomatoes and planting scene: (a) cherry tomatoes, (b) tomato growth environment, and (c) tomato planting facilities. (Units: mm). [Color figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com/doi/10.1002/rob.70250)]

TABLE 1 | Tomato harvesting robot requirements.

Design parameters	Request
Width of harvesting platform	≤ 0.8 m
Length of the robot	≤ 3 m
Height of harvesting	1100–1500 mm
Radius of harvesting by robotic arm	≥ 800 mm
Steering mode	In-place turning
Travelling form	Ground track dual purpose

Based on the growth height of tomatoes and the dimensions of the cultivation environment, design requirements for the tomato harvesting robot are specified in Table 1 to ensure maneuverability and harvesting efficiency. Given that the spacing between planting rack rows is only 1600 mm and the pedestrian area width is merely 3000 mm, the harvesting robot must be capable of in-place turning.

3 | Materials and Methods

3.1 | Hardware Design

3.1.1 | Overall Design

The tomato harvesting robot primarily consists of a mobile chassis (self-designed and manufactured), a lifting platform (self-designed and manufactured), a robotic arm (Aubo i5), an end-effector (self-designed and manufactured), two RGB-D cameras (Realsense L515), and a control system, as shown in Figure 2.

The mobile chassis propels the robot on both the ground and tracks, while supplying power to the robotic arm and controller. The chassis is capable of in-place turning; detailed information on the chassis design is provided in Section 1.1 of the

Supporting Information. The robotic arm drives the end-effector to perform harvesting and bin placement tasks. The end-effector is responsible for cutting and gripping the tomato stalks. The vision system primarily comprises a depth camera and recognition software, tasked with identifying and obtaining the spatial coordinates of the tomato stalks, and feeding this information back to the robotic arm. The control system manages the overall scheduling and control of the robot, deployed within an industrial personal computer (IPC), which is installed inside the mobile chassis.

As shown in Figure 3, all hardware is centered around the IPC, with controllers of other components interconnected to it. The IPC serves as the master controller, coordinating the mobile chassis, robotic arm, end-effector, and RGB-D camera through control software, enabling the robot's autonomous movement, recognition, harvesting, and bin placement operations.

Figure 4 illustrates a simplified layout of the tomato cultivation base. During operation, the tomato harvesting robot can automatically ascend and descend the track and switch between tracks. Upon mounting the track, the robot activates one side camera to identify ripe tomatoes. Once a harvestable tomato is detected, the robot halts its movement and re-localizes the spatial coordinates of all visible ripe tomato stems. The robotic arm then positions the end-effector at the stalk, where it performs cutting and gripping. The harvested tomato is temporarily stored in a crate. Subsequently, the robotic arm moves the end-effector to harvest the next visible tomato bunch, repeating the process until all visible ripe tomatoes are collected. Upon reaching the end of the track, the robot reverses along the track, automatically switches to the opposite camera, and continues harvesting until it returns to the starting point, at which point it automatically switches to the next row for continued harvesting.

3.1.2 | Design of End-Effector

To adapt to the complexities of the tomato harvesting environment, the end-effector structure needs to be compact and simple. Measurements indicate that tomato stems have

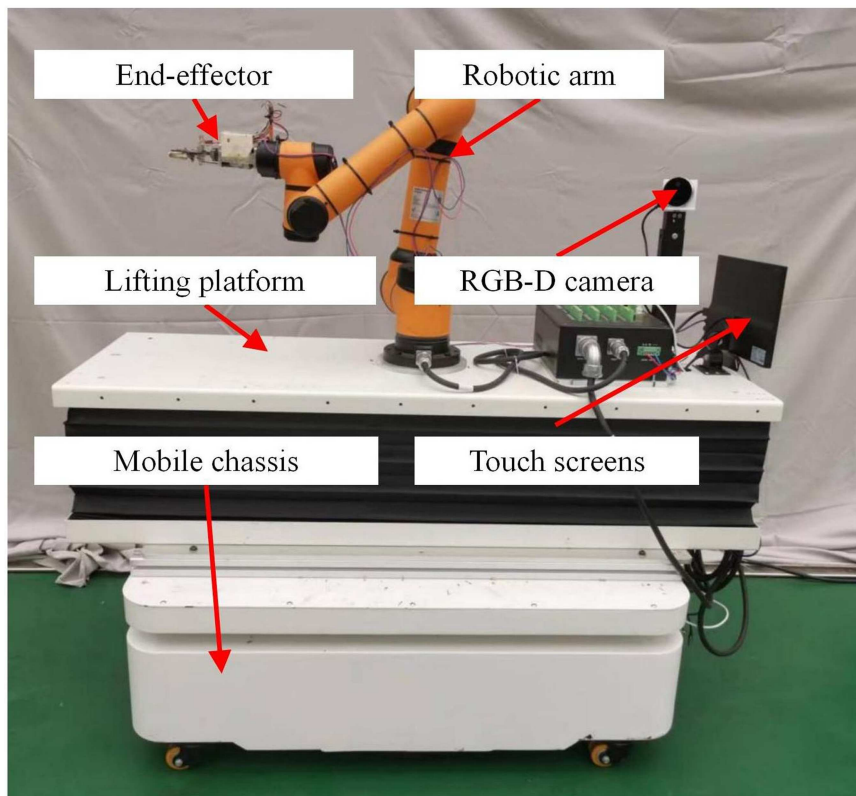


FIGURE 2 | Tomato harvesting robot prototype. [Color figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com)]

diameters ranging from 2.48 to 5.12 mm, and each tomato bunch weighs between 250 g and 500 g. The end-effector must be capable of cutting and clamping the stems to harvest the tomatoes.

To meet the design requirements, the cutting and clamping modules of the end-effector are integrated, as shown in Figure 5. This integrated design allows both cutting and clamping of the tomato stalk using a single power source. When the cutting module closes, a torsion spring transfers power to the clamping module, preventing damage to the stalk.

The end-effector primarily consists of blades, clamp blocks, torsion springs, rack, motor, nut, lead screw, support, connecting rods, and push rod. The motor drives the lead screw to rotate, causing the nut to move axially along the lead screw. The push rod, connected to the nut, moves back and forth, driving two connecting rods that form a parallelogram with the tails of the two blades, enabling the blades to perform cutting and opening actions.

Clamp block 2 is rigidly connected to blade 2 via bolts. During the cutting action, clamp block 2 moves in unison with blade 2, and the torsion spring drives clamp block 2 to engage with clamp block 1, performing the clamping action. Clamp block 1 features a slot. During stalk clamping, clamp blocks 1 and 2 contact the stalk and begin to close. If the stalk's reaction force exceeds the torsion spring's force, clamp block 1 ceases to tighten, preventing damage to the stalk. Meanwhile, blades 1 and 2 continue to approach each other, completing the cutting and clamping of the stalk.

After the end-effector holding the tomato stalk reaches above the fruit basket, the motor actuates the clamp blocks to release. As clamp blocks 1 and 2 separate, the tomato is released into the basket.

In this study, the contact surfaces between the blades, clamp blocks, and tomato stems are designed as concave curves, as illustrated in Figure 5c. If the blade-stalk contact surfaces are flat, the force analysis is depicted in Figure 5b, where F_1 and F_2 represent the pressures exerted by the right and left blades on the stalk, and f_1 and f_2 denote the corresponding friction forces. The analysis indicates that, regardless of the stalk's position, the resultant force F of F_1 and F_2 is directed toward the blade opening, while the combined friction force f of f_1 and f_2 is directed toward the blade hinge. If F exceeds f , the stalk tends to slide toward the blade opening, posing a risk of the stalk slipping out of the blade.

To mitigate this issue, the blades are designed with concave curves, as shown in Figure 5d. When the stalk is near the blade opening, as depicted in Figure 5d (i), the resultant force exerted by the blades on the stalk tends to move the stalk toward the hinge. If the resultant force F exceeds the friction force f , the stalk moves toward the hinge; if F is less than f , the stalk remains stationary.

When the stalk is positioned at the center of the blades, as illustrated in Figure 5d (ii), F_1 and F_2 balance each other, and the stalk remains stationary during the cutting process.

When the stalk is near the blade hinge, as shown in Figure 5d (iii), the stalk tends to move toward the blade opening. If the

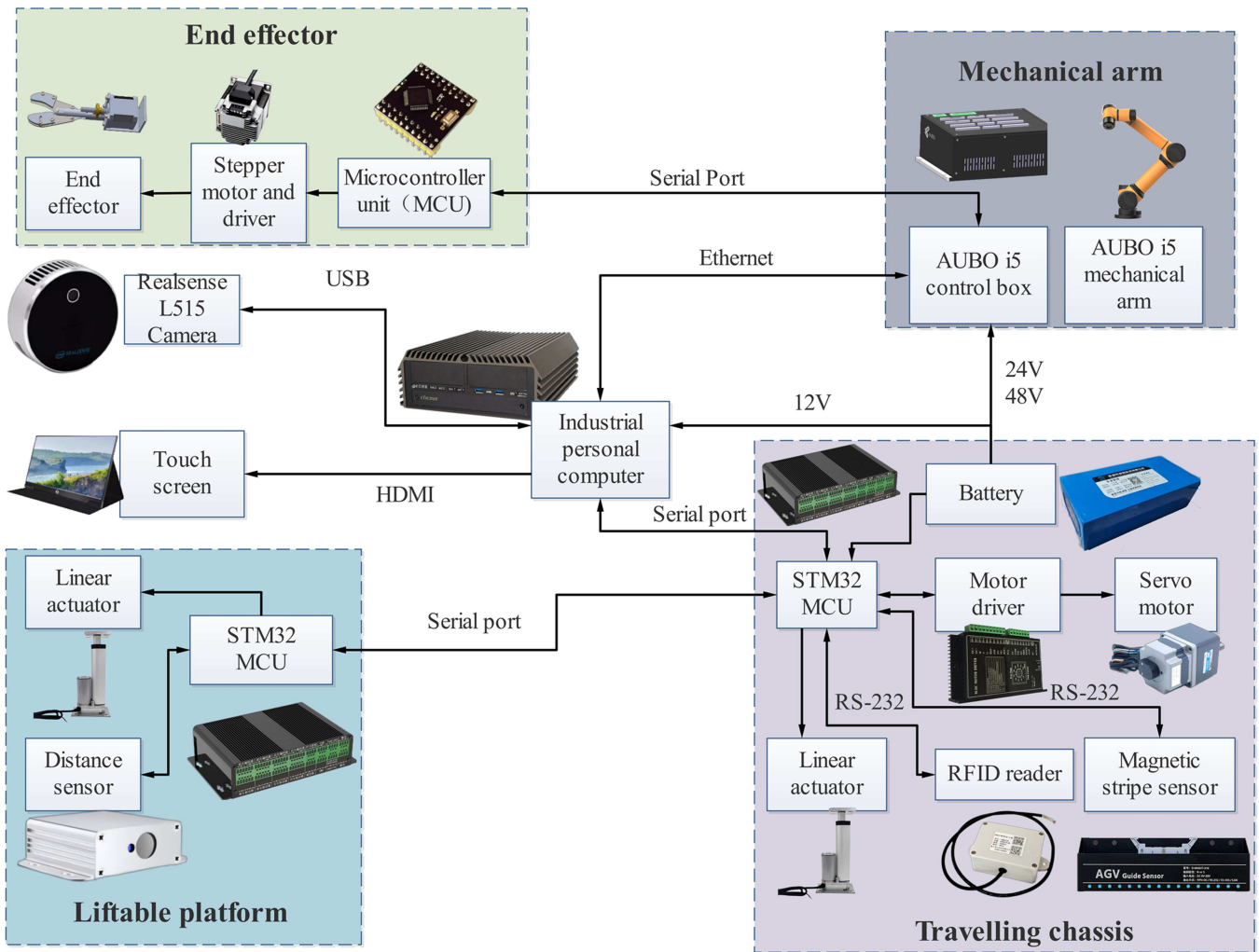


FIGURE 3 | Tomato harvesting robot hardware. [Color figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com)]

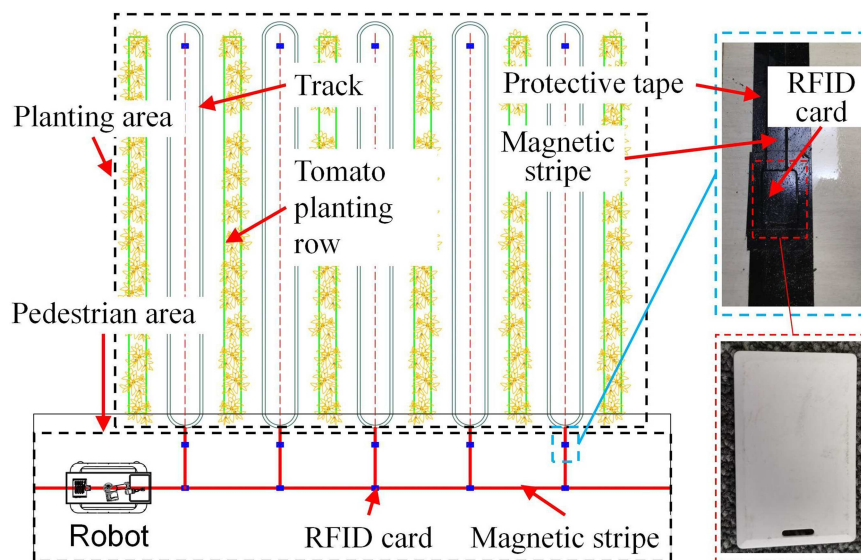


FIGURE 4 | Tomato harvesting robot traveling chassis. (Units: mm). [Color figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com)]

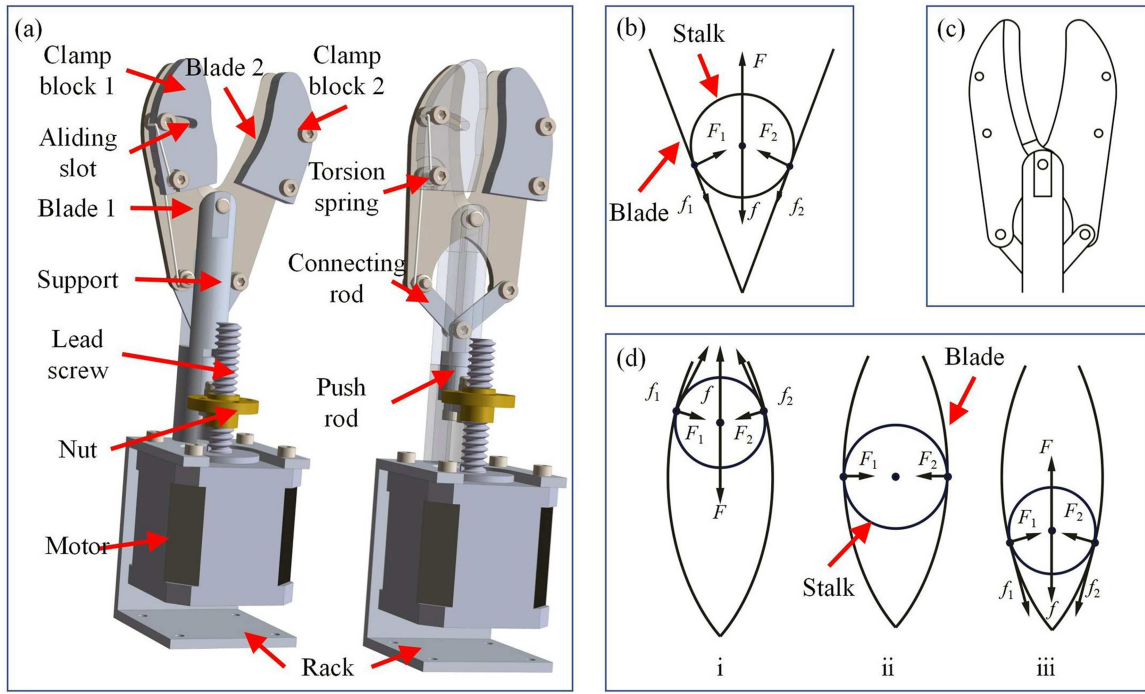


FIGURE 5 | End effector: (a) CAD model of the harvesting end effector, (b) force analysis of the straight blade, (c) CAD model of our blade, and (d) force analysis of our blade. [Color figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com)]

resultant force F exceeds the friction force f , the stalk moves toward the opening but does not slip out of the blades; if F is less than f , the stalk remains in place.

Based on the above analysis, compared to straight blades, the concave blade design ensures that as long as the stalk is within the blade range, it will not slip out, thereby preventing cutting failure.

3.1.3 | Camera Placement Design in Robotic Systems

The Intel RealSense L515 was selected as the vision sensor for recognition and localization, offering a color image field of view of $70^\circ \times 43^\circ$, a depth image field of view of $70^\circ \times 55^\circ$, and a depth sensing range from 0.25 m to 9 m. As shown in Figure 2, an eye-to-hand configuration was adopted, wherein the camera is fixed in the workspace and observes the robot's movements from a stationary position. To facilitate the harvesting of tomatoes on both sides of the track, two cameras were installed on either side of the robot's chassis, each responsible for recognizing tomatoes on their respective sides of the track. Upon entering a track, the robot activates the camera on one side to identify and harvest ripe tomatoes. After reaching the end of the track, it switches to the other camera to perform recognition and localization on the opposite side, thereby enabling the harvesting of tomatoes on both sides of the track.

To determine the appropriate distance and orientation between the camera and the robotic arm, a parameterized design approach was adopted. A schematic diagram illustrating the camera's placement and orientation is presented in Figure 6. It was ensured that the camera and the robotic arm do not interfere with each other; specifically, in the initial state, the robotic arm should not obstruct the camera's field of view, and

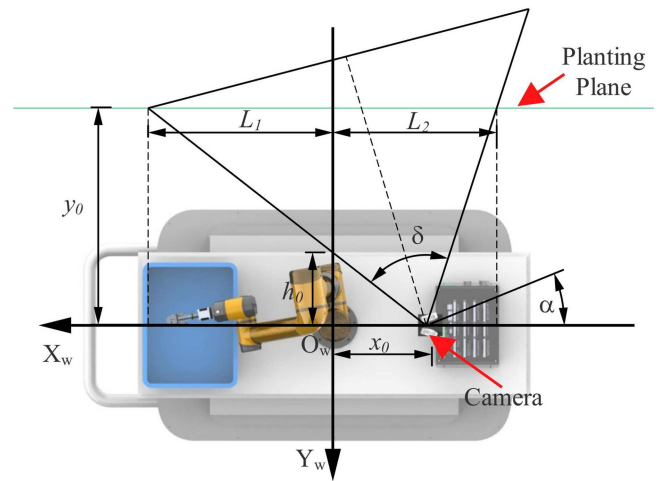


FIGURE 6 | Schematic of camera placement and stance. [Color figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com)]

the camera should not impede the movement of the robotic arm. Therefore, a certain distance was maintained between the camera and the base of the robotic arm. Given the distance between the camera and the robotic arm, if the camera is aligned parallel to the planting plane, the field of view limitations may result in capturing mostly tomatoes on the side of the arm closer to the camera. This could lead to identifying tomatoes located beyond the arm's reach, thereby underutilizing the arm's capabilities. To address this, the camera is tilted at a certain angle.

The coordinate system $X_w O_w Y_w$ represents the robotic arm base, where y_0 in Figure 6 denotes the distance between the arm's installation point and the planting plane, specified as

800 mm in this study. x_0 represents the distance between the camera and the robotic arm base. h_0 denotes the distance between the intersection line of the field of view plane and the $Y_W O_W Z_W$ plane of the robotic arm base coordinate system, and the $X_W O_W Z_W$ plane. α represents the tilt angle of the camera. δ denotes the field of view angle of the camera, which is 70° in this study. L_1 and L_2 represent the left and right field widths of the camera relative to the robotic arm.

In this application scenario, the camera's viewing angle should satisfy the following conditions: L_1 and L_2 should be approximately equal; the camera's tilt angle should be minimized; the camera's installation position should not impede the robotic arm's movement; and in the arm's initial state, it should not obstruct the camera's field of view. L_1 and L_2 can be calculated using Equation (1) and Equation (2), respectively, while h_0 can be determined using Equation (3).

$$L_1 = \frac{y_0}{\tan\left(90 - \frac{\delta}{2} - \alpha\right)} - x_0 \quad (1)$$

$$L_2 = \frac{y_0}{\tan\left(90 - \frac{\delta}{2} + \alpha\right)} + x_0 \quad (2)$$

$$h_0 = x_0 \tan\left(90 - \frac{\delta}{2} - \alpha\right) \quad (3)$$

To fully utilize the performance of the robotic arm, the left and right field widths of the camera should be equal. Let $L_1 = L_2 = L$. By substituting $y_0 = 800$ mm and $\delta = 70^\circ$ into Equation (1) and Equation (2), Equation (4) can be derived.

$$x_0 = \frac{400}{\tan(55 - \alpha)} - \frac{400}{\tan(55 + \alpha)} \quad (4)$$

Substituting Equation (4) into Equation (3) yields Equation (5):

$$h_0 = 400 - \frac{400}{\tan(55 - \alpha)\tan(55 + \alpha)} \quad (5)$$

Substituting Equation (4) into Equation (1), the left and right field widths L of the camera relative to the robotic arm can be expressed as Equation (6).

$$L = \frac{800}{\tan(55 - \alpha)} - \left(\frac{400}{\tan(55 - \alpha)} - \frac{400}{\tan(55 + \alpha)} \right) \quad (6)$$

The curves of Equation (4) and Equation (6) are plotted in Figure 7.

To ensure equal left and right field widths of the camera relative to the robotic arm, as the camera angle α increases, both the distance x_0 between the camera and the robotic arm and the field width L increase. However, a larger α leads to greater discrepancies between the pose of the stalk in the camera coordinate system and that in the robotic arm coordinate system. Therefore, α should be minimized, and consequently, a smaller x_0 is preferable.

To prevent the camera from interfering with the robotic arm's movement and to avoid the arm obstructing the camera's field of view in its initial state, the Auboi5 robotic arm has a joint distance of 223 mm from the base in its initial position. Thus, h_0 should be greater than or equal to 223 mm. Substituting $h_0 = 223$ mm into Equation (4) yields $\alpha = 13.88^\circ$, which is rounded up to 14° .

Substituting $\alpha = 14^\circ$ into Equation (4) results in $x_0 = 306.60$ mm, which is rounded to 310 mm. With $\alpha = 14^\circ$ and $x_0 = 310$ mm, the calculated h_0 is 269.48 mm, ensuring that the camera's field of view is not obstructed by the robotic arm. The field widths L_1 and L_2 are 610.29 mm and 617.09 mm, respectively, indicating a minimal difference. Additionally, the camera installation height is set to 400 mm.

3.2 | Robot System Design

3.2.1 | Software System Architecture

The tomato harvesting robot primarily comprises a chassis, a robotic arm, an end-effector, two cameras, a lifting platform,

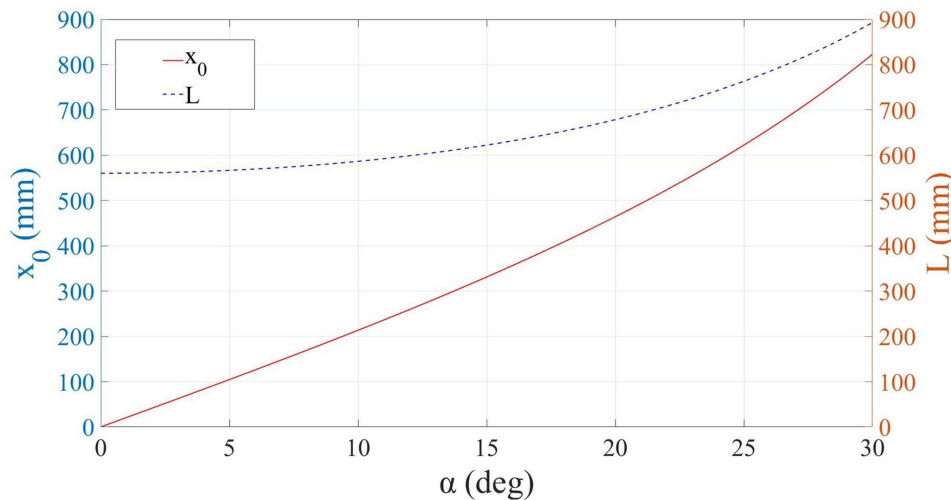


FIGURE 7 | Distance, viewing width, and camera angle relationship. [Color figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com)]

and an industrial computer, with the hardware connections illustrated in Figure 3.

A host control system and human-machine interface (HMI) for the tomato harvesting robot were developed on the industrial computer using Qt, with the overall control program architecture depicted in Figure 8.

To coordinate the control of various components, the control system is divided into several modules: chassis motion control, navigation, user interface (UI), main program, robotic arm and end-effector control, lifting platform height control, and visual recognition.

The main program serves as the core, responsible for coordinating and scheduling other modules, processing data received from them, and issuing control commands accordingly.

The chassis motion control module primarily handles serial communication, sending motion commands to the chassis and receiving status information, which is then forwarded to the navigation module.

The navigation module reads the route map and, based on it and the chassis status, sequentially calculates and issues motion control commands to the chassis motion control module; it also processes movement or stop commands from the main program and uploads chassis status data back to it. To enable autonomous navigation covering all harvesting rows, the system employs a partitioned navigation method that integrates magnetic stripes, RFID cards, and track constraints, allowing the robot to automatically switch between the Pedestrian area and the Planting area (Supporting Information, Figures S2 and S3). Detailed descriptions of the navigation system design and navigation path planning are provided in Sections 1.2 and 1.3 of the Supporting Information.

The UI module facilitates parameter settings and data retrieval for the operator.

The lifting platform control module sends the platform height data set via the UI to the platform controller; the height is configured before robot operation and does not participate in the harvesting motion control.

The robotic arm and end-effector control module generate motion paths and end-effector states based on the harvesting point coordinates from the main program and sequentially send them to the robotic arm controller.

The visual recognition module processes the color and depth images captured by the camera to generate harvesting point coordinates, which are then uploaded to the main program.

The main program runs on the main thread, while the navigation and visual recognition modules operate on separate sub-threads, ensuring non-blocking execution of the main program.

3.2.2 | Control Flow

An event-triggered approach was proposed to coordinate chassis motion and harvesting, and a main program was developed as the core of the control system, with the control flow depicted in Figure 9. The blue boxes represent the navigation module, visual recognition module, robotic arm, and end-effector control module, which interact with the main program through internal communication.

Initially, all modules are initialized, and the robot moves according to the navigation route map. When the visual recognition module detects a harvestable tomato, it commands the chassis to stop moving.

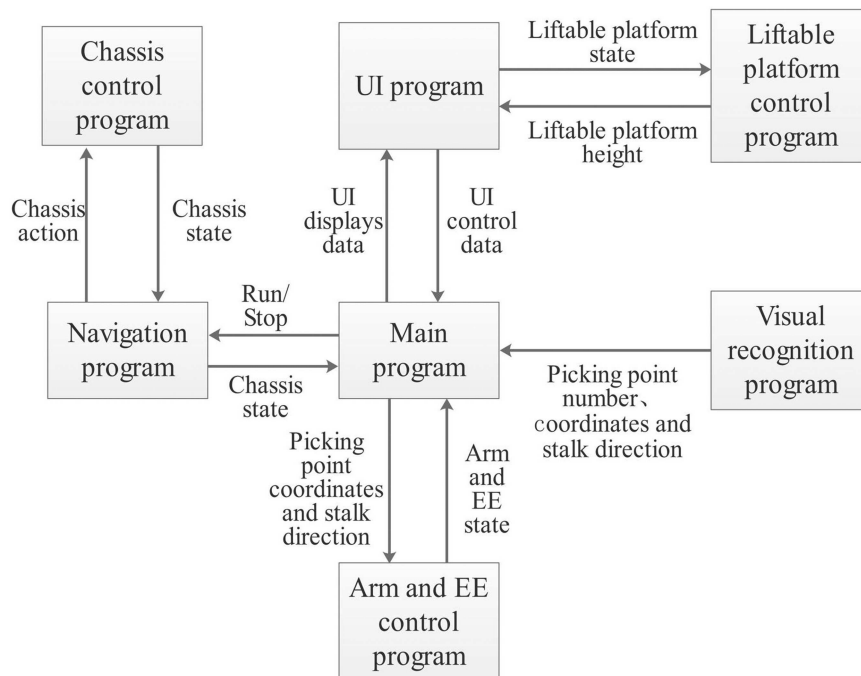


FIGURE 8 | Overall control program architecture.

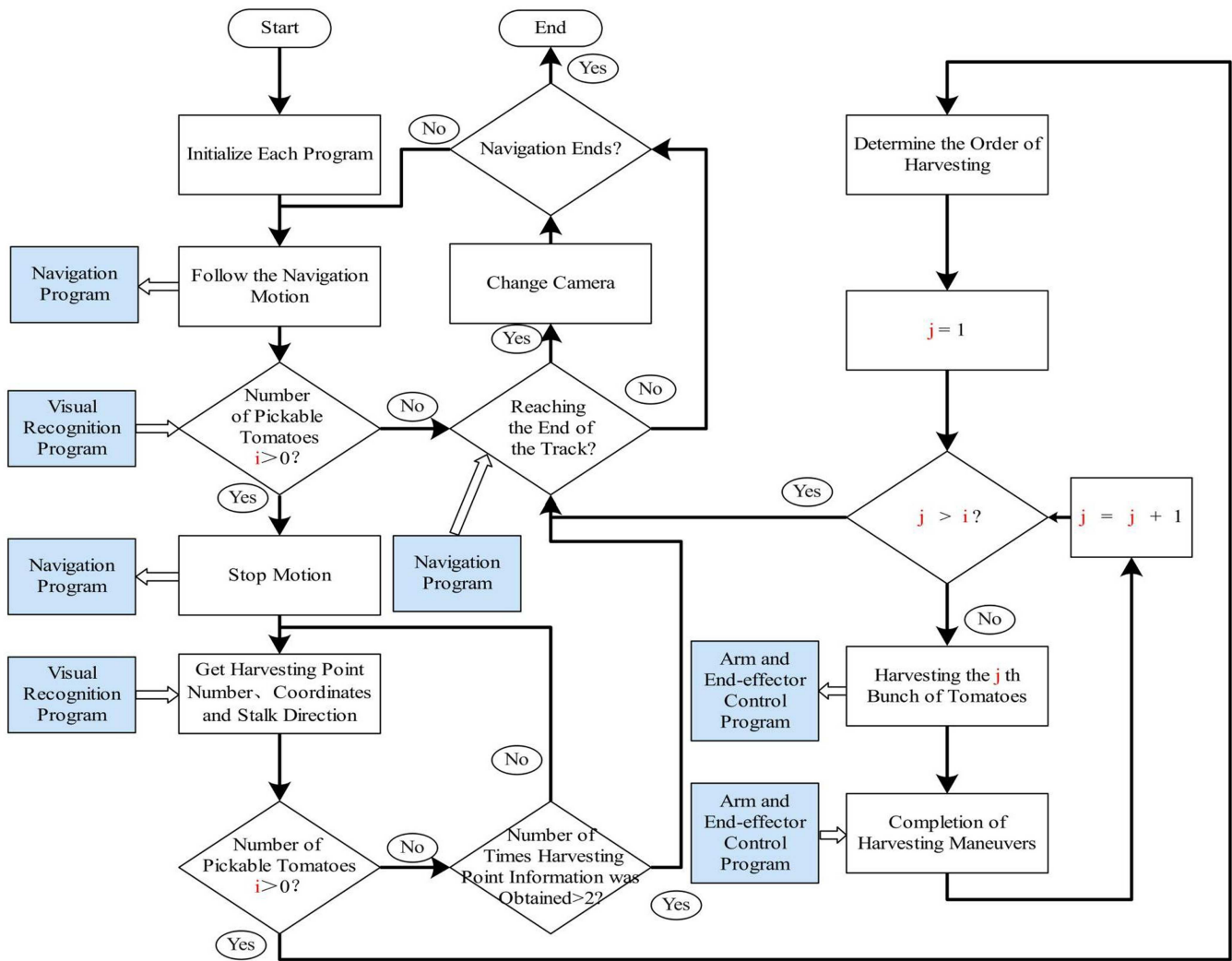


FIGURE 9 | Control flow chart. [Color figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com)]

After the chassis stops, the visual recognition module retrieves the number and spatial coordinates of the harvesting points. Since the camera captures higher-quality color and depth images in a stationary state, leading to more accurate recognition, if the coordinates cannot be obtained in one attempt, it will try again. If both attempts fail, the robot continues moving forward.

If the harvesting point coordinates are obtained in both attempts, the harvesting sequence is determined based on the stalk's coordinates. To avoid interference from other tomato bunches, the stalk's harvesting point coordinates in the robotic arm coordinate system are first sorted by the X-coordinate in descending order, then by the Z-coordinate in descending order.

According to the computed harvesting sequence, the harvesting point coordinates and stalk direction of the first tomato bunch are sent to the robotic arm and end-effector control programs, after which the robotic arm performs path planning and harvesting. To avoid collisions between the robotic arm and the plants, a three-point robotic arm path planning method was developed, which is detailed in Section 2 of the Supporting Information. After the first bunch is harvested, the coordinates

of the second bunch are sent, and so on, until all detected tomatoes are harvested.

After harvesting all tomatoes within the field of view, if the robot reaches the end of the track, it switches the camera to harvest tomatoes on the other side; if not, it continues moving forward. The above steps are repeated until all harvesting areas defined in the navigation route map are completed.

3.3 | Perception System

3.3.1 | Harvesting Point Detection and Localization

In 2020, Alexey Bochkovskiy introduced the YOLOv4 algorithm for object detection (Bochkovskiy et al. 2020). The YOLO series of algorithms has since been widely applied by researchers for the identification of agricultural crops such as cassava stalks and tomatoes (Singhpoo et al. 2023; Chen et al. 2024; Zhang et al. 2024a; Zheng et al. 2022), demonstrating the feasibility of using this method for recognizing tomato bunches and stalks. Our research team previously proposed a YOLOv4-based algorithm for recognizing tomato bunches and stalks (Qin et al. 2021), which was successfully applied to the detection and localization of the stalks in Jinlinglong, Israeli red variety, and Yuekeda202 tomatoes. These

three tomato varieties exhibit orange or red coloration when ripe. To improve the generalization performance of the recognition algorithm, the Yellow variety A44YG was added to the training dataset, enabling the identification of orange, red, and yellow tomatoes.

First, the trained YOLOv4 model is used to identify and locate tomato bunches and stalks. Then, traditional image processing methods are applied to the stalk region to obtain the pixel coordinates of the harvesting points. Finally, the spatial coordinates of the harvesting points in the camera coordinate system are calculated using the depth map provided by the camera.

The YOLOv4 recognition results are shown in Figure 10a, where mature tomato bunches and stalks are enclosed in bounding boxes. Each output box from YOLOv4 contains position, class, and confidence information, allowing for post-processing to determine the posture of the mature tomato stalks and the locations of the harvesting points.

Based on the stalk prediction information, the stalk regions are segmented from the color and depth images. A depth threshold is applied to segment the background in the color image.

In Chinese greenhouse tomato cultivation, the typical row spacing ranges from 1400 to 1800 mm. The harvesting robot developed in this study is designed to navigate between crop rows, with the camera positioned centrally to simultaneously identify fruits on both sides. Consequently, the horizontal distance between the camera and the harvestable tomatoes is approximately 700 to 900 mm. To enhance the accuracy of background removal and mitigate the impact of depth measurement errors on recognition results, a safety margin was incorporated, and the depth threshold was ultimately set to 1000 mm.

After background segmentation of the stalk region in the color image using the depth threshold, methods such as k-means clustering, morphological operations, and the Zhang-Suen algorithm for thinning are applied sequentially to process the stalk color image, obtaining the refined stalk region. The processing flow is shown in Figure 10e–h (Qin et al. 2021).

Then, based on the stalk growth direction (described in Section 3.3–3), the coordinates of the stalk harvesting points are obtained, as shown in Figure 10i–k. If the stalk grows to the left or right, the intersection of a vertical line with the refined stalk region is taken as the pixel coordinates of the harvesting point, as shown in

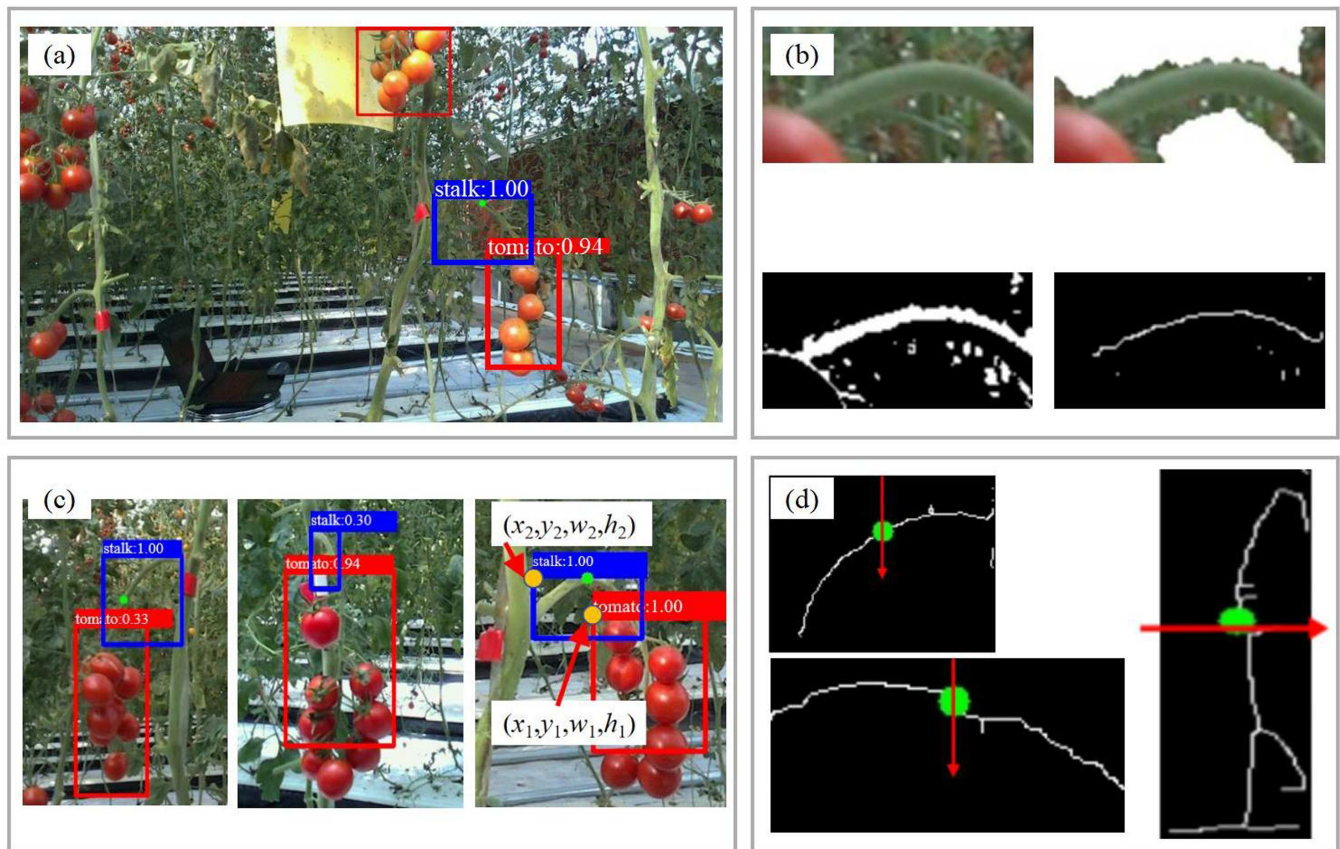


FIGURE 10 | Tomato harvesting point recognition: (a) detection result demonstration based on YOLOv4, (b) stalk thinning process: original stalk segmentation → background removal based on depth threshold → stalk extraction via K-means clustering → morphological operations and thinning, (c) Schematic of three tomato-bunch growth orientations: left-facing, front-facing, and right-facing, (x_1, y_1, w_1, h_1) denotes the pixel coordinates (x_1, y_1) of the top-left corner and the width w_1 and height h_1 of the tomato bunch ROI; (x_2, y_2, w_2, h_2) denotes the pixel coordinates (x_2, y_2) of the top-left corner and the width w_2 and height h_2 of the stalk ROI, (d) method for determining the harvesting point under different growth orientations: for left or right facing stalks, the harvesting point is the intersection of the detection box's vertical center line with the refined stalk region; for front-facing stalks, the harvesting point is the intersection of the detection box's horizontal center line with the refined stalk region. [Color figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com)]

Figure 10i,j; if the stalk grows forward, the intersection of a horizontal line with the refined stalk region is taken as the pixel coordinates of the harvesting point, as shown in Figure 10k.

The depth value of the point corresponding to the harvesting point pixel coordinates is obtained from the depth map. Using this depth value and the camera intrinsic parameters, the spatial coordinates of the harvesting point in the camera coordinate system (${}^{obj}T_{cam}$) are calculated. The pose matrix between the camera and the robotic arm base (${}^{cam}T_{base}$), obtained using a nine-point calibration method, is then used to compute the spatial coordinates of the harvesting point in the robotic arm coordinate system (${}^{obj}T_{base}$); detailed descriptions of the hand-eye calibration method are provided in Section 3 of the Supporting Information. The Euclidean distance L from the harvesting point to the robot base is computed. If L exceeds the robot's operational range, the harvesting point data is discarded; if L is within the operational range, the spatial coordinates and stalk growth direction of the point are saved.

The above steps are repeated until all identified stalk spatial coordinates (${}^{obj}T_{base}$) within the field of view are processed. The stalk spatial coordinates and growth direction information are sent to the main program for its use. The visual recognition system operates as a child thread and does not interfere with the main program's operation.

3.3.2 | Tomato Bunch and Stalk Matching Method

As the recognition results encompass all tomatoes and stalks, and the identified tomatoes and stalks in the image do not exhibit a one-to-one correspondence (as shown in Figure 10a), it is necessary to pair tomatoes with their corresponding stalks to facilitate subsequent identification of tomato stalk growth postures.

In our previous research, tomatoes whose region of interest (ROI) overlapped with those of the stalks were considered as matching tomatoes for the respective stalks, forming a tomato bunch (Qin et al. 2021). However, due to variations in tomato growth states and limitations in deep learning recognition precision, the ROI of stalks and tomato bunches within a bunch often do not overlap, resulting in a low recall rate for matching stalk and bunch ROI.

In this study, the stalk and tomato bunch ROI with the closest proximity are matched as a pair. If a tomato bunch ROI intersects with a given stalk ROI, the tomato bunch with the largest intersection ratio is selected to pair with the stalk. If no tomato bunch ROI intersects with the stalk ROI, it is determined whether there exists a tomato bunch ROI within a certain distance threshold from the stalk ROI, and whether a single parallel translation can result in an intersection. If such a tomato bunch ROI exists, the closest one is selected to pair with the stalk.

Based on experimental results, the threshold is set to 15% of the smaller dimension between the image width and height (refer to Section 4.2 for details).

3.3.3 | Stalk Growth-Direction Qualitative Discrimination Method

The growth postures of tomato stalks vary, as illustrated in Figure 10c. In our earlier work, based on statistical data of stalk

growth postures and cutting analysis of the end-effector, for tomato bunches growing to the left and to the right, the end-effector, during harvesting, is rotated by 30° around the Y_w -axis in the corresponding direction, while for tomatoes growing forward, the end-effector remains horizontal without rotation (Zou 2023). This method enables effective harvesting of tomatoes, since it does not require precise identification of stalk posture, thereby reducing the load on the perception system.

As shown in Figure 11, an angle β was defined between the tangent at the tomato stalk harvest point and the normal of the harvesting plane and was measured manually. When β was between 0° and 30°, the growth direction was classified as forward. When β ranged from 30° to 150°, the growth was classified as lateral, with further distinction into left-growing or right-growing depending on the stalk's position relative to the stem. When β was between 150° and 180°, the growth direction was defined as backward.

As shown in Figure 10(c), the aspect ratio of the fruit stalk region of interest (ROI) was defined as $\lambda = w_2/h_2$, where w_2 and h_2 denote the width and height of its bounding box, respectively. This ratio varied with growth direction, being smaller for forward-growing stalks than for lateral-growing stalks. Therefore, a threshold on λ was applied to distinguish forward-growing from sideways-growing stalks.

The pseudocodes for the stalk growth-direction qualitative discrimination method are presented in Algorithm 1. Initially, the relative position between the centers of the fruit and stalk ROIs is used to preliminarily classify the growth as leftward or rightward. Then, the final growth direction is determined based on the λ threshold. Due to the camera's 14° tilt, the thresholds for distinguishing left and right stalk growth differ. Based on experiments, the λ threshold is set to 0.6 for leftward growth and 0.9 for rightward growth (refer to Section 4.4 for details)

ALGORITHM 1 | Determination of tomato stalk growth direction

Require: Match the ROI coordinates of the tomatoes and stalk. (x_1, y_1, w_1, h_1) and (x_2, y_2, w_2, h_2)

Ensure: Stalk growth direction. (*Left/Forward/Right*)

1: **if** $(x_1 + \frac{w_1}{2}) < (x_2 + \frac{w_2}{2})$ **then**

2: **if** $\frac{w_2}{h_2} > 0.6$ **then**

3: $p \leftarrow \text{Left}$

4: **else**

5: $p \leftarrow \text{Forward}$

6: **end if**

7: **else**

8: **if** $\frac{w_2}{h_2} > 0.9$ **then**

9: $p \leftarrow \text{Right}$

10: **else**

11: $p \leftarrow \text{Forward}$

12: **end if**

13: **end if**

14: **return** p

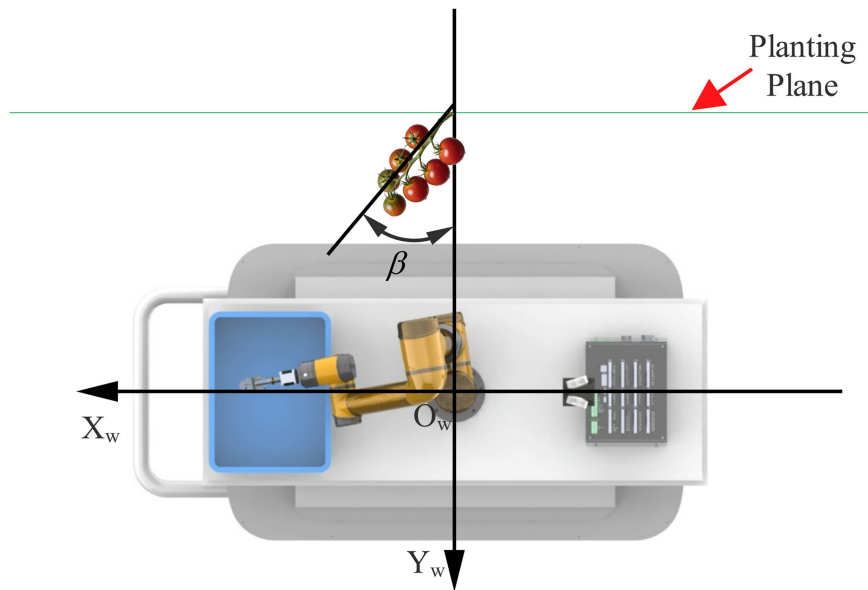


FIGURE 11 | Schematic of stalk growth orientation. [Color figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com/doi/10.1002/rob.70250)]

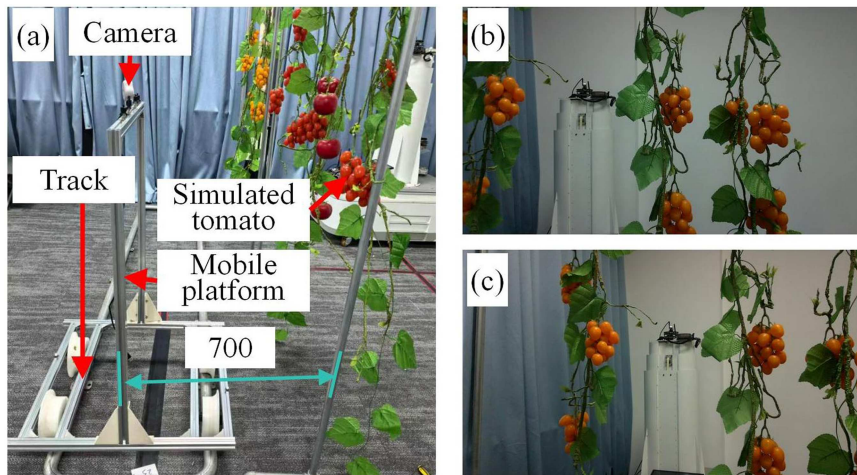


FIGURE 12 | Tomato simulation scene and captured images. (a) Tomato harvesting simulation platform. Example image captured when the camera shooting plane is fixed at (b) 0° and (c) 14° to the direction of travel. [Color figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com/terms-and-conditions)]

4 | Results and Discussion

4.1 | Parameterized Camera Design Verification Experiment

To validate the effectiveness of the parameterized camera design described in Section 3.3-3, a laboratory experiment was performed using a simulated tomato harvesting scenario (Figure 12a). Simulated tomato bunches were suspended on a rack by vines, while a mobile platform carrying the camera moved along a track. The rack was positioned 700 mm from the platform's centerline, and the camera was mounted on the platform with adjustable viewing angles. Two configurations were tested: 0° and 14° relative to the motion direction. Starting from the end of the track, the platform advanced in 25 cm increments, and at each position, one color image and one depth image were recorded, yielding 11 image pairs per configuration.

From each color-depth pair, the spatial coordinates of any pixel in the color image were computed within the camera coordinate system. Based on the spatial relationship between the camera and the robotic arm base (Section 3.3-3), the relative pose transformation was established, enabling the determination of these coordinates in the robotic arm base coordinate system. The Aubo-i5 robotic arm has a moving-range radius of 784 mm at point P, and with an end-effector length of 160 mm, the maximum reachable stalk distance was 944 mm from the base origin. For both tested camera angles (0° and 14°), the distances of all detected stalks to the base origin were measured, with distances below 944 mm defined as harvestable.

The experimental results demonstrated that the harvestable proportion of tomatoes was 61.90% (39/63) at a camera angle of 0° and increased to 78.33% (47/60) at a camera angle of 14° . This represents an improvement of 16.43% compared with the 0°

configuration, thereby confirming the effectiveness of the proposed parameterized camera design.

4.2 | Visual Recognition Experiment

The identification of tomato bunches and stalks is the first step in tomato harvesting; thus, the precision of this recognition determines the overall success rate of subsequent harvesting operations. A total of 6622 images of tomatoes were collected under varying lighting conditions, including 2953 images of the Jinlinglong variety, 1321 images of the Israeli variety, 1073 images of the Yuekeda202 variety, and 1275 images of the Yellow Variety A44YG, as shown in Figure 13. All 6627 images were annotated accordingly. From the collected images, a random selection of 336 Jinlinglong, 147 Israeli, 46 Yuekeda202, and 76 Yellow Variety A44YG tomato images was designated as the test set to evaluate the performance of the YOLOv4 recognition model. All image data used in this study were collected in real greenhouse environments, covering various times from morning to evening and diverse shooting angles to ensure data diversity.

To enhance the generalization capability of the training model, data augmentation techniques were applied to the remaining 6022 tomato images (excluding the test set), and the augmented data were used for training. The training was conducted on a computer running Ubuntu, equipped with two Intel Xeon Silver 4210 CPUs, two Nvidia GeForce RTX 2080 Ti GPUs, and 128 GB of RAM. The optimal YOLOv4 model achieved a mean Average Precision (mAP) of 79.55%, and this model was selected as the final YOLOv4 training result. To evaluate the performance of the trained model, the aforementioned test set was used for recognition using this model.

As shown in Table 2, the precision and recall rates for stalks or tomato bunches of each variety exceeded 80.00%. The overall precision and recall rates for tomato bunches were 85.04% and 88.71%, respectively, while those for stalks were 82.72% and 81.51%, respectively. The precision and recall rates are comparable to those reported by (Zheng et al. 2024; Rong et al. 2024). However, this study's model can recognize

multiple tomato varieties (Jinlinglong, Israeli red variety, Yuekeda202, and Yellow Variety A44YG), demonstrating better generalization. Future research can expand the tomato variety database to further enhance the model's generalization capability, thereby facilitating adaptation to different tomato varieties and promoting commercial applications.

In tomato bunch recognition, false positives (FPs) primarily consist of immature tomatoes (48.44%), mature tomatoes from non-harvesting rows (39.06%), and background elements (12.50%). The presence of immature tomatoes among FPs in tomato bunches may be attributed to the annotation process, where tomatoes with over 80% color change were labeled as mature. This threshold, determined subjectively, may include some green tomatoes, leading the recognition model to misclassify certain immature tomatoes as mature bunches. Future research could involve identifying the color of each tomato and determining the proportion of color-changed tomatoes within a bunch to assess its maturity.

The occurrence of mature tomatoes from non-harvesting rows among FPs in tomato bunches may be due to two factors. The first reason could be the variation in camera angles during data collection using different devices (e.g., smartphones, DJI Pocket 2, RealSense L515, and RealSense D435i) to augment the dataset, resulting in perspectives differing from those during actual harvesting. The second reason might be manual annotation errors, where tomatoes from non-harvesting rows or distant harvesting rows were mistakenly labeled as mature, causing the model to misidentify some non-harvesting row mature tomatoes as part of mature bunches. If such misidentifications occur, depth maps can be utilized to apply threshold segmentation, filtering out tomatoes from non-harvesting rows.

4.3 | Tomato Bunch and Stalk Matching Experiment

After identifying the tomatoes and stalks, matching is required. As described in Section 3.3-2, stalks and tomato bunches with

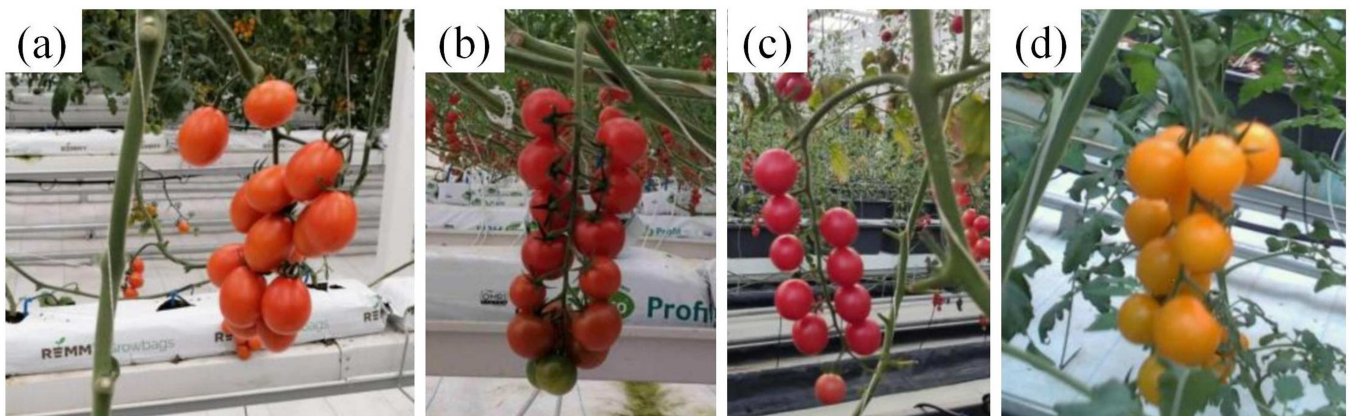
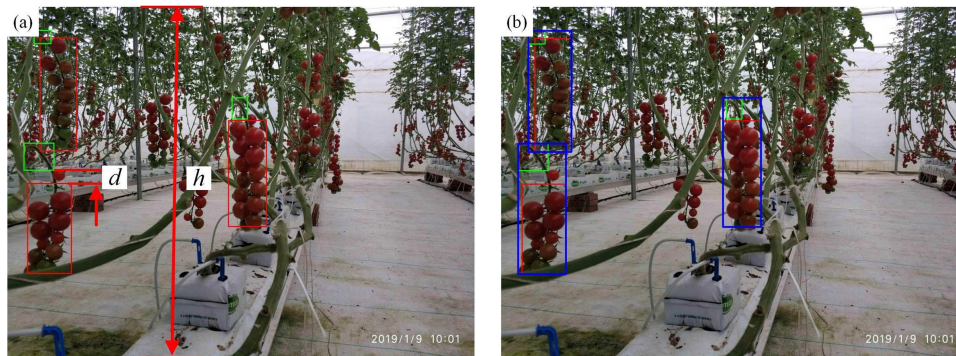


FIGURE 13 | Tomato: (a) Jinlinglong, (b) Israeli red variety, (c) Yuekeda202, and (d) Yellow Variety A44YG. [Color figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com)]

TABLE 2 | Recognition results.

Variety Region	Jinlinglong		Israeli Red Variety		Yuekeda202		Yellow Variety A44YG	
	Tomato bunch	Stalk	Tomato bunch	Stalk	Tomato bunch	Stalk	Tomato bunch	Stalk
TP	584	384	207	163	107	83	131	93
FP	97	82	32	32	22	15	30	22
FN	72	90	27	33	13	17	19	24
Precision	85.76%	82.40%	86.61%	83.59%	82.95%	84.69%	81.37%	80.87%
Recall	89.02%	81.01%	88.46%	83.16%	89.17%	83.00%	87.33%	79.49%
F1	0.87	0.82	0.88	0.83	0.86	0.84	0.84	0.80

**FIGURE 14** | Matching tomato bunches with their stalk: (a) before matching, (b) after matching. [Color figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com)]

similar ROI distances are matched to form a tomato bunch. This study conducts experiments using the ratio of ROI distance (d) to image height (h), defined as μ , to analyze the impact of different μ values on the recall and precision of stalk and tomato bunch matching. A total of 800 images were collected, with 100 randomly selected as the training set and the remaining 700 as the test set. Matching was performed at levels of 0%, 5%, 10%, 15%, 20%, and 25%, and the recognition results were observed. Some examples are shown in Figure 14, and the experimental results are presented in Table 3.

As shown in Table 3, when μ varies from 0% to 25%, recall first increases and then decreases; precision remains constant initially and then decreases. When μ is 15%, recall is 99.07% and precision is 100.00%, yielding the highest recall and precision. Therefore, μ is set to 15% for subsequent tomato and stalk matching in this study. To validate the generalizability of this threshold, tomato and stalk matching was performed on the 700 test images with $\mu=15\%$, achieving a recall of 99.22% and a precision of 100.00%.

4.4 | Stalk Growth-Direction Qualitative Discrimination Method Experiment

As described in Section 3.3-3, to enhance the success rate of tomato harvesting, it is necessary to perform Qualitative Discrimination of the stalk's growth-direction. This study employs the relative positional relationship between the stalk ROI and the tomato bunch ROI, along with the aspect ratio of the stalk ROI, to ambiguously determine the growth orientation of the stalk.

TABLE 3 | The matching results of tomatoes and stalks of the training data to identify the optimal μ .

μ	Recall	Precision
0%	53.70%	100.00%
5%	93.52%	100.00%
10%	96.30%	100.00%
15%	99.07%	100.00%
20%	96.30%	99.02%
25%	94.44%	97.14%

To determine the threshold value of λ used for assessing the stalk's growth orientation, this study collected 500 images using a tomato harvesting robot's camera, encompassing stalks growing forward, leftward, and rightward. The growth directions were manually annotated. A random selection of 100 images constituted the training set, while the remaining 400 images formed the test set.

In the randomly selected training set, a total of 113 identifiable stalks were present. Among these, 32 stalks grew forward, with 17 having the fruit on the left side and 15 on the right side; 44 stalks grew leftward, and 37 grew rightward.

The λ values were set to different values to recognize the stalk's growth direction, and the corresponding true positive rate (TPR) values were obtained and plotted as shown in Figure 15. Here, λ_L and λ_R denote the values when the tomato bunch is located on the left and right sides of the stalk, respectively.

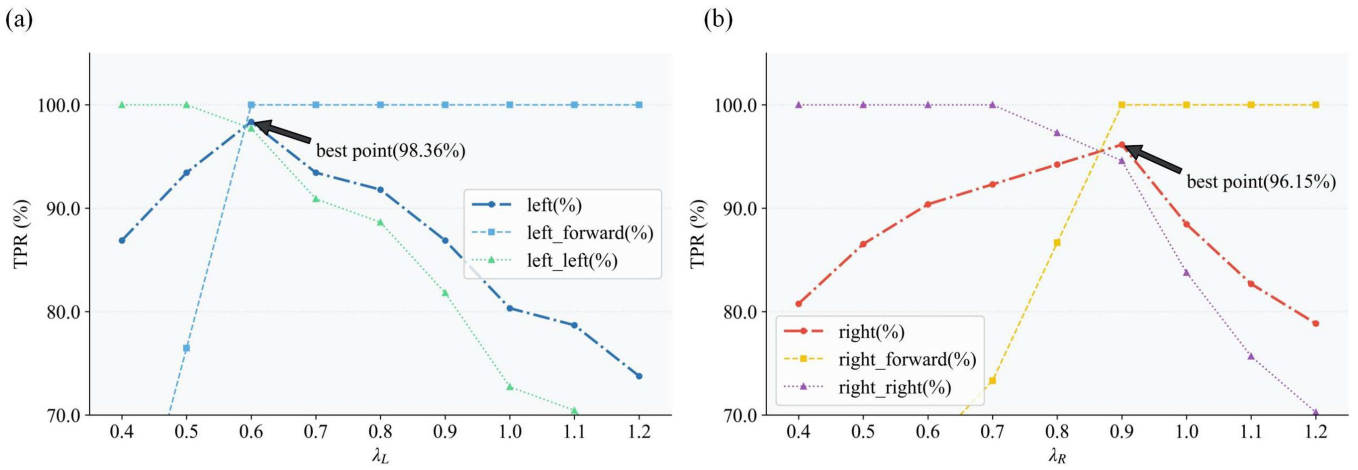


FIGURE 15 | Impact of the value of λ on TPR, λ_L and λ_R represent the values when the tomato bunch is located on the left and right sides of the stalk, respectively: (a) impact of the value of λ_L on TPR, the left_forward curve illustrates the relationship between TPR and the λ value when the tomato bunch is situated on the left side of the stalk, and the stalk grows forward; the left_left curve depicts the relationship between TPR and λ when the tomato bunch is on the left side of the stalk, and the stalk grows leftward; the left curve represents the overall relationship between TPR and λ for cases where the tomato bunch is on the left side of the stalk, (b) impact of the value of λ_R on TPR, the right_forward, right_right, and right curves are defined in the same manner as described above. [Color figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com)]

As observed from Figure 15, the highest TPR was obtained when $\lambda_L = 0.6$ and $\lambda_R = 0.9$. Accordingly, subsequent studies adopted $\lambda = 0.6$ for cases where the fruit was on the left side of the stalk and $\lambda = 0.9$ for cases where it was on the right.

To validate the generalization capability of the above thresholds, these values were applied to the test set. The TPR achieved was 97.28% when the fruit was on the left side of the stalk and 97.59% when on the right side.

4.5 | End-Effector Cutting and Gripping Experiments

To evaluate the gripping and cutting performance of the end-effector, separate experiments were conducted for gripping and cutting.

4.5.1 | End-Effector Gripping Experiment

The end-effector was divided into three regions: near the opening, the middle section, and near the hinge, to test the gripping force at each region.

To assess the impact of different end-effector orientations on gripping performance, the angle (γ) between the cutting plane of the end-effector and the bunch stalk was set to 90°, 60°, and 30°, respectively, as shown in Figure 16.

The end-effector gripped the tomato stalk, and a handheld digital push-pull force meter (manufacturer: Weidu Electronics, model: WDF-50, accuracy: $\pm 1\%$) was used to apply a pulling force at the end of the stalk, as shown in Figure 16, until the stalk slipped from the end-effector. The maximum force recorded represents the gravitational force corresponding to the maximum tomato weight the end-effector can grip.

As shown in Figure 17a, when the gripping position remains constant, a smaller γ angle results in a greater gripping force;

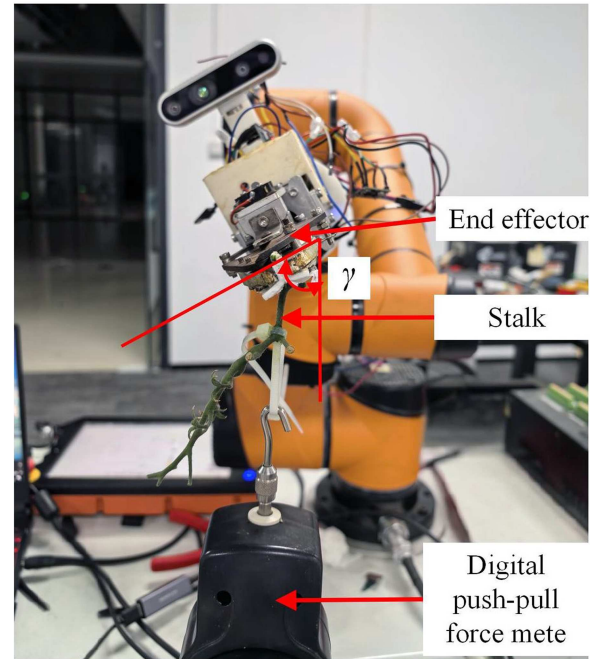


FIGURE 16 | Gripping and shearing experiment of the end-effector. [Color figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com)]

when γ is constant, the gripping force at the opening and hinge areas is less than that at the middle section.

The minimum gripping force of the end-effector was 16.02 N, sufficient to grip a 1600 g tomato, which exceeds the maximum bunch weight of 500 g recorded by our team.

4.5.2 | End-Effector Cutting Experiment

To evaluate the influence of various end-effector postures and cutting positions on the effectiveness of stalk cutting, experiments were conducted using different cutting regions of the

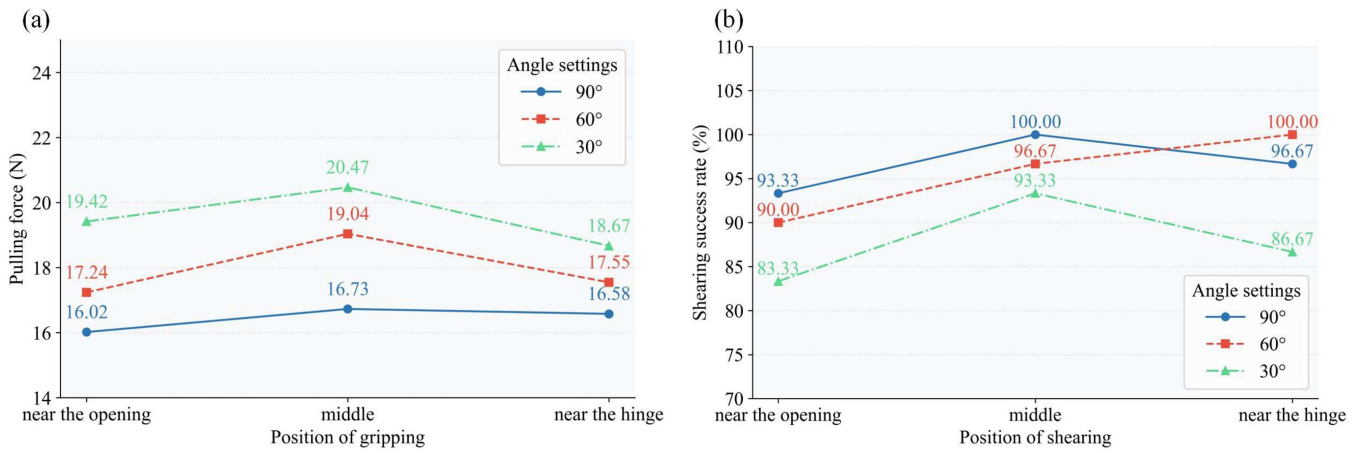


FIGURE 17 | Results of the performance test of the end-effector: (a) Impact of the position of gripping on force, (b) Impact of the value of the position of shearing on success rate. [Color figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com)]

end-effector (near the opening, the middle section, and near the hinge) and varying angles between the cutting surface and the stalk (90°, 60°, and 30°). Each combination was tested on 30 tomato stalks, and the cutting success rate was calculated.

As illustrated in Figure 17b, the overall cutting success rate across all combinations of cutting positions and angles was 93.33%. For a fixed cutting position, a smaller angle γ resulted in a lower cutting success rate; conversely, for a fixed γ angle, cutting at the middle and hinge regions of the end-effector yielded higher success rates compared to the near the opening region. Specifically, the lowest success rate of 83.33% was observed when cutting near the opening region with a γ angle of 60°.

These findings suggest that positioning the stalk closer to the middle or hinge regions of the end-effector enhances cutting effectiveness. To mitigate the impact of recognition errors and ensure higher success rates during actual tomato harvesting, it is recommended to align the tomato stalks nearer to the hinge region of the end-effector.

4.6 | Tomato Harvesting Experiment

The tomato harvesting robot for bunch harvesting was developed by integrating the aforementioned technologies, with its key technical parameters presented in Table 4. After constructing the bunch tomato harvesting robot, field experiments were conducted in a tomato garden at a research institute in Guangzhou, Guangdong Province, China. In accordance with agronomic practices, defoliation was performed prior to tomato ripening, ensuring minimal leaf obstruction for the harvestable tomatoes. Before the harvesting trials, all harvestable tomatoes in the designated rows were manually marked with tape, without affecting their detection and harvesting. During the robot's operation, human intervention in the harvesting process was avoided unless necessary to prevent potential damage to the tomato plants.

The tomato harvesting robot moved forward along a track, harvesting tomatoes on one side. Upon detecting a harvestable tomato, the robot halted to perform the harvesting operation.

After harvesting, it resumed forward movement until reaching the end of the track. Subsequently, the robot reversed along the track, harvesting tomatoes on the opposite side, continuing until it returned to the starting position, thereby completing the harvesting trial for the row.

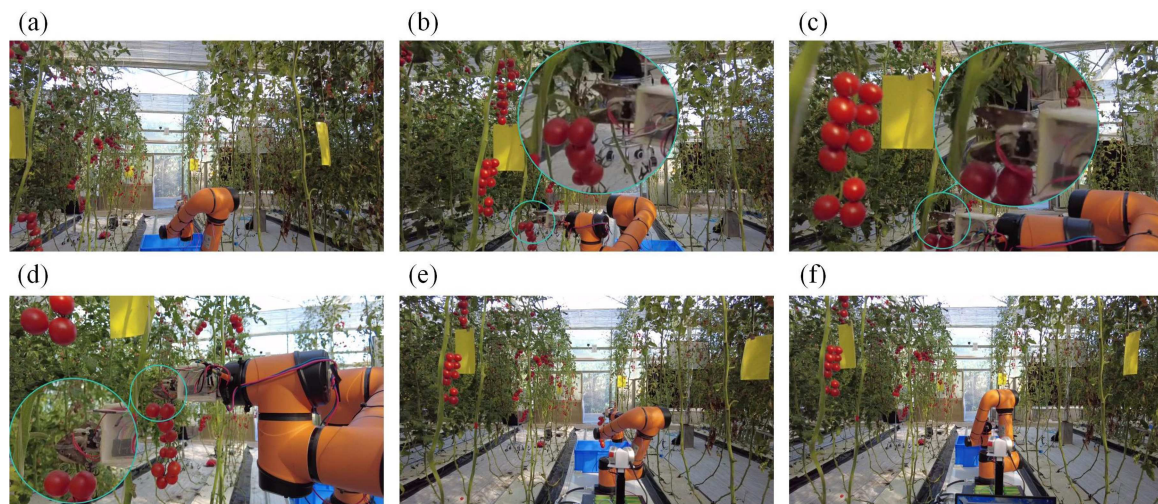
As illustrated in Figure 18, upon obtaining the spatial coordinates of the harvestable tomato stalk, the robotic arm guided the end-effector from its initial state to a position in front of the stalk's harvesting point with an appropriate orientation, as shown in Figure 18b. Subsequently, the end-effector advanced to the precise harvesting point, as depicted in Figure 18c. Upon reaching the harvesting point, the end-effector executed a closing action to simultaneously cut and grasp the stalk, effectively detaching the bunch from the stalk, as illustrated in Figure 18d. The robotic arm then transported the tomato to a position above the collection basket, as shown in Figure 18e. Once positioned above the basket, the tomato was released into it, completing the harvesting operation.

A total of 65 tomato bunches were manually marked during the trial, among which 46 were successfully harvested, resulting in a harvesting success rate of 70.77%. A successful harvest was defined as the accurate localization of the stalk harvest point by vision recognition, followed by stalk cutting and detachment with the end-effector and placement of the tomato into the basket by the robotic arm. The average harvesting time per bunch was 12.23 s. The remaining 19 failures were systematically analyzed and classified into three major categories with eight sub-categories, as summarized in Table 5.

The most significant failure factor was the perception system (57.90%), which comprised four sub-categories. First, harvestable fruit stalk were not recognized (31.58%): the recognition model achieved only 81.51% recall for stalks, resulting in some being misclassified as background. Second, stalks were obscured (10.53%): in some camera images, stalks were occluded by the main stem or adjacent bunches, leading to recognition failures. Third, stalk-to-fruit matching failure (10.53%): tomatoes at the top of the bunch were over-ripened and detached, which prevented the bunch and its corresponding stalk from being matched using the Tomato bunch and stalk

TABLE 4 | The key parameters of the robot.

No	Component	Manufacturer	Model	Parameter	Specification
1	Robotic arm	Aubo, China	aubo-i5	(1) Payload (kg)	5.00
				(2) Reach (mm)	886.50
				(3) Repeatability (mm)	± 0.02
2	Cameras	Intel, USA	Realsense L515	(1) Ideal range (m)	0.25–9.00
				(2) Depth accuracy (mm)	5.00–14.00
				(3) Depth field of view (°)	70 × 55
				(4) RGB sensor field of view (°)	70 × 43
3	IPC	Decheng, China	DS-1302	(1) CPU	i5-10500
				(2) GPU	RTX-2060
				(3) RAM (G)	8
4	Mobile chassis	Self-fabricated	Self-fabricated	(1) Size (m)	1.30 × 0.84 × 0.49
				(2) Speed (m/s)	0–1.20
				(3) Steering mode	Pivot turn
				(4) Number of magnetic stripe sensors	2
5	End-effector	Self-fabricated	Self-fabricated	(1) Size (m)	0.172 × 0.04 × 0.05
				(2) Payload (kg)	1.60
				(3) Cutting success rate (%)	83.08–100.00
6	Harvesting robot	/	/	(1) Harvesting success rate (%)	70.77
				(2) Efficiency (s/bunch)	12.23
				(3) Size (m)	1.40 × 0.84
				(4) Working height range (m)	0.50–2.00

**FIGURE 18** | Tomato harvesting procedure: (a) initial state, (b) reaching the insertion point (point C), (c) reaching the harvesting point (point B), (d) end-effector closure for shear separation of tomato stalks, (e) transportation over fruit baskets, and (f) release the tomato into the basket and restore it to its original state. [Color figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com)]

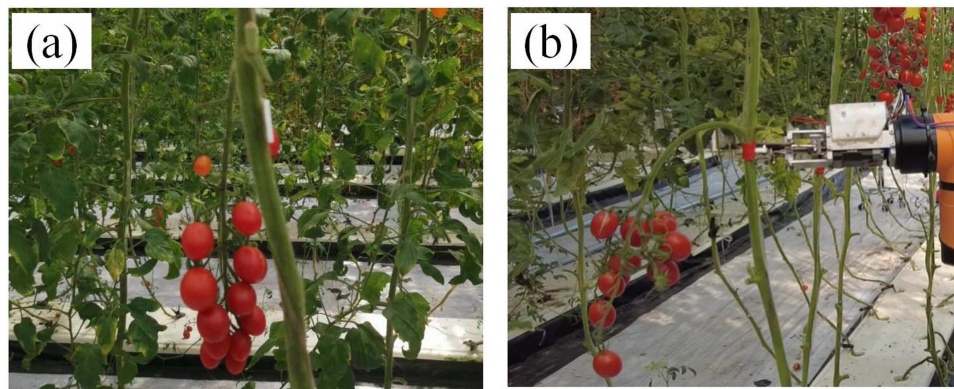
matching method (Section 3.3-2). Fourth, missing depth values (5.26%): with stalk diameters of only 2.48–5.12 mm, the depth camera provided insufficient point density, leaving the harvest point and its surroundings without depth values and preventing spatial coordinate computation.

The second major source of failure was the execution system (31.58%), which comprised three sub-categories. First, stalks

growing toward the rear of the stem (21.05%): although the angled camera could still identify and localize the harvest point (Figure 19a), the robotic arm approached from the front and moved parallel to the Y_W axis (Figure 6). In this process, the harvest point was occluded by the stem, and the end-effector often interfered with or even enclosed the stem within the scissors (Figure 19b), resulting in harvesting failure. Second, collisions with non-harvested fruits (5.26%): during robotic arm

TABLE 5 | Summary of robot harvesting failures.

Categories	Failure reasons	Failure times	Incidence in failures(%)
Issue of perception system	Harvestable fruit stalk not recognized	6	31.58
	Stalk obscured by stem or bunch	2	10.53
	Stalk and bunch matching failed	2	10.53
	Missing depth value	1	5.26
Issue of execution system	Obstructed by the main stem	4	21.05
	Obstructed by the other tomato bunch	1	5.26
	End-effector shearing failure	1	5.26
Others	Cumulative error in stalk harvesting point localization	2	10.53

**FIGURE 19** | Examples of failed harvesting: (a) tomatoes growing toward the rear, (b) the end-effector is obscured by the main stem when harvesting tomatoes growing toward the rear. [Color figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com)]]

motion, impacts with surrounding fruits altered the coordinates of the target stalk harvest point, leading to unsuccessful cutting. Third, end-effector shearing failure (5.26%): relatively thick stalks positioned at the opening of the end-effector were not fully severed, causing harvesting failure.

The remaining failures were attributed to cumulative errors (10.53%). Specifically, inaccuracies arising from recognition, camera depth values, hand-eye calibration, and robotic arm motion accumulated over the process, ultimately preventing the end-effector from accurately reaching the stalk harvest point.

5 | Conclusions

Due to the significant labor costs associated with manual tomato harvesting, which constitute a major portion of overall production expenses, this study proposes a tomato harvesting robot designed to automate the harvesting process and reduce associated costs.

This paper presents the design and development of a tomato harvesting robot, integrating components such as a mobile chassis, robotic arm, end-effector, camera system, and control unit to achieve automation in functions including navigation, tomato detection, stalk cutting, bunch separation, and placement into baskets, based on the coordinated design of the chassis structure and the navigation system, the robot can

cruise and harvest continuously triggered by the vision system, enabling long-term and uninterrupted operation. Field trials were conducted in a real-world environment to evaluate its performance.

An end-effector was designed specifically for harvesting tomato bunches, utilizing a single power source to perform both cutting and gripping actions. The cutting blade was shaped with a concave surface to prevent tomatoes from slipping during the cutting process.

The installation position and orientation of the camera were analyzed to ensure that the target tomatoes were within the camera's field of view and to prevent interference between the camera and the robotic arm.

A YOLOv4-based method was employed for the detection of stalks and tomatoes, with subsequent matching of stalks to bunches. Additionally, a qualitative discrimination method was proposed to determine the growth direction of tomato bunches.

Finally, experiments were conducted to validate the parameterized camera design, recognition methods, matching algorithms, stalk growth-direction qualitative discrimination method, end-effector performance, and harvesting operations in real-world scenarios.

Parameterized Camera Design Verification Experiment.

TABLE 6 | Performance or functional parameters.

Researcher and affiliation	Recognition precision	Recall rate	Harvesting success rate	Harvesting efficiency
Zheng et al. (2024)	Tomato fruit: 82.62%	Tomato fruit: 82.57%	57.50%	14.90 s/bunch
Rong et al. (2024)	Tomato bunch: 98.7% and 100%	Tomato bunch: 71.96% and 66.18%	57.70% and 55.4%	24.00 s/bunch
Li et al. (2024)	Tomato bunch maturity: 89.71%	/	86.67% (Including only tomatoes growing forwards or rightwards)	32.46 s/bunch
This paper	Tomato bunch: 85.04%; Stalk: 82.72%	Tomato bunch: 88.71%; Stalk: 81.51%	70.77%	12.23 s/bunch

The integration of various modules resulted in a prototype of the tomato harvesting robot, which underwent field trials. The visual recognition model achieved an overall precision and recall of 85.04% and 88.71% for tomato bunches, respectively, and 82.72% and 81.51% for stalks, respectively. The recall rate for matching stalks and bunches is 99.22%; the qualitative discrimination for stalk growth direction exceeds 97%; the overall cutting success rate of the end-effector is 93.33%, with a gripping capacity of 1600 g. The harvesting success rate of the tomato harvesting robot reached 70.77%, with an average successful harvest time of 12.23 s, indicating that the robot can generally meet the practical harvesting requirements.

We conducted a structured comparison with the latest studies, and the performance parameters are summarized in Table 6. The harvesting experiment environment in this study was comparable to those reported by Zheng et al. (Zheng et al. 2024) and Rong et al. (Rong et al. 2024), under which our robot achieved a higher harvesting success rate. Although the system developed by Li et al. (Li et al. 2024) reported an even higher success rate, its applicability is limited because only forward-growing and right-growing tomatoes were considered in the evaluation. In contrast, the tomato harvesting robot developed in this study further supports autonomous traversal, enabling movement along the greenhouse track without human intervention and thereby substantially enhancing operational efficiency.

Regarding the limitations of this study, future work should focus on more accurate recognition of stalk growth orientations and obtaining the relative pose between the stalk and the main stem.

Secondly, adding an additional camera with an installation angle parallel to the harvesting plane can help avoid misjudgment of stalk growth directions.

The current harvesting operation is executed in an open-loop manner, without feedback on harvesting success, which may cause unintended cutting of the main stem and consequent economic losses. To address this limitation, future improvements will incorporate a feedback mechanism in which the camera detects the diameter of plant material entering the end-effector cutting area. Owing to the pronounced diameter difference between the main stem and the tomato stalk, this feature can be exploited to reliably distinguish whether the object entering the cutting zone is a stalk or the main stem.

Finally, optimizing the end-effector's posture based on stalk growth directions can facilitate harvesting of both backward and forward-growing stalks, thereby improving the success rate.

Acknowledgments

This work was supported by the Fruit and Vegetable Harvesting Robot Development under Grant No. Yue Cai [2023] 6; the Fundamental Research Funds of the Central Universities under Grant No. 2024ZYGXZR028; the Department of Agriculture and Rural Affairs of Guangdong Province Modern Agricultural Industry Common Key

Technology R&D Innovation Team Project of Guangdong Province (Facility and Equipment, Planting Direction) under Grant No. 2026CXTD01; the Guangdong Basic and Applied Basic Research Foundation under Grant No. 2026A151010118; and the Guangdong S&T Program under Grant No. 2026B0101110003.

Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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Supporting Information

Additional supporting information can be found online in the Supporting Information section.

Supporting File: rob70250-sup-0001-Supporting_InformationV3.rar.